

EEF Meeting Minutes, Thursday July 20, 2023

Erlang Ecosystem Foundation, Minutes of a Regular Meeting of the Board of Directors Meeting #84, July 20, 2023

A regular meeting of the Erlang Ecosystem Foundation was held over group conference call (Zoom). The meeting began at 16:06 UTC.

Attendance

The following members of the board of directors were present (11/11): Francesco Cesarini, Sophie DeBenedetto, Maxim Fedorov, Amos King, Kenneth Lundin, Ben Marx, Bryan Paxton, Miriam Pena Villanueva, Peer Stritzinger, Sebastian Strollo, and Alistair Woodman.

The following members of the board of directors were absent: none.

Agenda Items

1. Approval of minutes from previous meetings.

RESOLVED, that the Erlang Ecosystem Foundation approve the minutes as presented as representing a true and accurate record of the board meeting #83 held on June 20, 2023. Unanimously approved.

2. Intermediate actions and activities: none
3. Treasury and Membership update: none
4. Unanimous consent votes: none scheduled.
5. Stipend requests: ARM32 JIT support proposal. Peer explained the background and answered questions about the proposal. Since the original proposal (attached) was submitted discussion has been made with the OTP team on how to make it an optional / externally supported module and there is an agreement on how this will be done. It is understood that this is a research project, without completing the work it is impossible to know what performance improvements one can expect from this work, but there is enough confidence that it will actually benefit the ARM32 platforms, in the form of speed improvements and possibly power savings, that the project seems worthwhile (and there is interest for this support from several users of ARM32 hardware). Another point of discussion was regarding the continued

support of the project: there is general agreement that the board should consider the “maintenance phase” (i.e. who will maintain a project once it is completed) when considering stipend requests, but also that there must be some kind of viability for a project to live on with the support of the community in the long run. The board was therefore reluctant to make a blanket approval of any ongoing costs to maintain the ARM32 JIT implementation at this point. Clarification from Peer regarding the requested amounts: the amounts requested are “not to exceed” (i.e. the time to implement is an estimate, if less time is required it will cost less, but if more time is required the EEF will not pay more than the requested amount).

RESOLVED, that the Erlang Ecosystem Foundation approve the request for funding of the ARM32 JIT proposal with the following modifications: only the first two steps, “Base Implementation” (at €30,000) and “Optimization” (at €10,000) are approved at this point, the amounts are a budget “not to exceed”. Vote count: 8 for (Sophie, Maxim, Amos, Ben, Bryan, Miriam, Sebastian, Alistair), 2 abstentions (Peer, Francesco), 1 against (Kenneth, who would like to see a lower amount).

6. Any other topics: none

7. Alistair declared the meeting adjourned at 17:05 UTC.

Dated July 20, 2024

Secretary of the meeting: _____ Sebastian Strollo

Attachments:

1. JIT support for ARM32 stipend requests (3 pages)

Stipend Proposal

Stipend Type : Development Work

First name : Peer

Last Name : Stritzinger

Email Address : peer@stritzinger.com

Organization Name : Peer Stritzinger GmbH

Country : Germany

Twitter : peerstr

Linkedin : <https://www.linkedin.com/in/peerst>

Website : www.stritzinger.com

Intended product of project :

For of OTP with working and optimized JIT for ARM32 architecture

Purpose

ARM32 is the most important embedded platform relevant to Erlang and Elixir. Almost all users of Nerves, meta-erlang and GRiSP will benefit

Development Plan

Currently the Erlang Runtime has a Just in Time compiler (JIT) for Intel architecture and ARM64. The underlying library for generating assembler code ASMJIT initially only supported Intel architecture and EEFF funded for it to be extended for ARM64 and ARM32. The ARM64 support was used by the OTP team to implement the JIT for this architecture. ARM32 is currently mainly used for Embedded Systems and IoT hardware and will be important in this space for a long time. In the Erlang Ecosystem there are a few embedded system stacks in use: Nerves (Erlang runtime on Buildroot Linux), meta-erlang (Erlang runtime on Yocto Linux) and GRiSP (Erlang runtime directly on hardware). A large majority of systems using these three stacks are based on ARM32 architecture and can't benefit currently from a JIT.

Unfortunately ARM64 and ARM32 are two completely different architectures and therefore need completely separate JIT implementations.

We wanted to tackle the ARM32 JIT support already last year (but were ultimately too busy to start it) and discussed the integration with the OTP team already back then. Since the OTP team doesn't want to get into the same situation with HiPE which was a drag on progress in the runtime because of ongoing maintenance efforts to keep it up to date. Therefore it was agreed that the Embedded WG would fork OTP and maintain the ARM32 JIT on this fork.

The project would work in three phases:

Base implementation. Starting with OTP26 all BEAM opcodes need to be implemented by the JIT. Once this is done the JIT would work on arbitrary BEAM code but not be necessarily faster. This is the largest junk of work.

Optimizations, based on type information passed down into the BEAM file by the compiler the generated assembly code will be optimized. Many runtime checks can be removed when the type, including possible value ranges is known. This is the source of the actual speedup provided by the JIT

Adaption to OTP27 when it comes out. Every major version can add new op-codes (often also for the benefit of JIT optimizations) which must be added to the JIT for it to stay usable. In addition it is to be expected that more type based optimization information is added to the BEAM files to allow generation of faster code.

After successful implementation of the ARM32 JIT for OTP27 yearly maintenance work needs to be done to keep the JIT useable. Ideally some effort should be also planned not only for the new opcodes but also to implement the new optimizations that will be dropping which each OTP version.

This is a substantial project which can not easily tackled by a individual, therefore we want to solve this with our whole team. We will do the internal accounting according to the same rules as for EU Framework Program (like Horizon Europe currently) i.e. we track the hours and calculate the hourly rates according to the salary of the employee + social security costs (simplified). Since there is a bit of flexibility in the scope (how much to optimize) this will be a guideline, but we expect to overdeliver a bit here.

Since the Nerves community and especially companies will be the most benefitting of the ARM32 JIT we can use this as a good argument to win them over to get EEF sponsor. So its very likely that the cost of the ongoing maintenance will be recovered by these new sponsors joining. I will personally work on facilitating that.

Stipend amount:

Base implementation: 30,000.— €

Optimisation: 10,000.—€

Adaption to OTP27 + further Optimisation: 10,000.—

Expected yearly cost for maintenance: 10,000.— (more to be known after we finished step 3).

Payment: 20,000.—€ at project start; 10,000.—€ after finishing Phase 1, further Phases after finishing.

Dissemination : We will do at least three blog posts and will submit at least one conference talk. Maybe also a paper for ICFP Erlang Workshop. We will especially target the Nerves community and will try to win new sponsors there.

Delivery Method : Github repo in the EEF org with forked OTP and ARM32 JIT added on a branch probably for each major OTP version a separate branch. As far as possible

CI integration (depends on progress on CI for OTP) We will create a project website probably using Github pages

Stipend Amount (USD) : 53500

Payment Method : Wire